5.1 We only show that μ_{12} is a measure in $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2)$ since μ_{21} can be showed with a similar manner. It suffices to prove the countable additivity of μ_{12}. Given a disjoint sequence $\{A_n\}_{n=1}^{\infty}$ in $\mathcal{F}_1 \times \mathcal{F}_2$. Let $B = \cup_{n\geq 1} A_n$, then by definition,

$$\mu_{12}(B) = \int_{\Omega_1} \mu_2(B_{1\omega_1}) d\mu_1(\omega_1),$$

where $B_{1\omega_1} = \{\omega_2 \in \Omega_2 \mid (\omega_1, \omega_2) \in B\}$. Clearly, for each fixed ω_1, $B_{1\omega_1} = \cup_{n\geq 1}(A_n)_{1\omega_1}$, where $\{(A_n)_{1\omega_1}\}_{n\geq 1}$ is a sequence of disjoint sets in $\mathcal{F}_2$ by Proposition 5.1.1. Now $\mu_2(B_{1\omega_1}) = \sum_{n\geq 1} \mu_2((A_n)_{1\omega_1})$, and notice that those $\mu_2(B_{1\omega_1}), \mu_2((A_n)_{1\omega_1})$ are all measurable functions who are valid due to the fact that μ_1, μ_2 are finite measures. **MCT** says $\int_{\Omega_1} \mu_2(B_{1\omega_1}) d\mu_1(\omega_1) = \sum_{n\geq 1} \int_{\Omega_1} \mu_2((A_n)_{1\omega_1}) d\mu_1(\omega_1)$, that is, $\mu_{12}(B) = \sum_{n\geq 1} \mu_{12}(A_n)$. Therefore, μ_{12} is a measure on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2)$. Similarly, μ_{21} is a measure on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2)$.

5.6 Use the set up in the hint from the book. Clearly $g = I(f(\omega) - t \geq 0)$ is $\mathcal{F} \times \mathcal{B}(\mathbb{R}^+)$-measurable, since $f(\omega)$ and t are both $(\mathcal{F} \times \mathcal{B}(\mathbb{R}^+), \mathcal{B}(\mathbb{R}))$-measurable functions. Notice that $\int_\Omega g(\omega, t) d\mu = \mu(\{f \geq t\})$ and $\int_{\mathbb{R}^+} g(\omega, t) dt = f$. Apply Tonelli's Theorem (all conditions there are satisfied, including σ-finiteness and nonnegative measurability), we get $\int_\Omega f d\mu = \int_{\mathbb{R}^+} \mu(\{f \geq t\}) dt$.

5.10 From $I = \int_0^\infty e^{-x^2/2} dx = \int_0^\infty e^{-y^2/2} dy$, we have

$$\begin{aligned}
I^2 &= \int_0^\infty e^{-x^2/2} dx \int_0^\infty e^{-y^2/2} dy \quad \text{(Tonelli's theorem)} \\
&= \int_0^\infty \int_0^\infty e^{-(x^2+y^2)/2} dx dy \quad (x = r\cos\theta,\ y = r\sin\theta) \\
&= \int_0^{\frac{\pi}{2}} \int_0^\infty e^{-\frac{r^2}{2}} r dr d\theta \quad \text{(Tonelli's theorem)} \\
&= \int_0^\infty r e^{-\frac{r^2}{2}} dr \int_0^{\frac{\pi}{2}} 1 d\theta \\
&= \frac{\pi}{2}
\end{aligned}$$

So $I = \sqrt{\frac{\pi}{2}}$.

5.11 The result is true when $\|f\|_1$ or $\|g\|_1$ are allowed to be infinity since the Tonelli's theorem is still true when the integral in either (2.3) or (2.4) is infinity. So we use those stuff in the hint. Notice that $h(x_1, x_2) \geq 0$ on $\mathbb{R}^2$, $\int_{\mathbb{R}^2} h(x_1, x_2) d\mu^2 \geq 0$, then the result follows immediately after applying Tonelli's Theorem.

In terms of random variables, this says, for nondecreasing functions g, h, $Cov(h(X), g(X)) \geq 0$ for any random variable X such that $E|h(X)|, E|g(X)| < \infty$.

A. (i) Let $\mathcal{C} = \{A \times B \mid A \in \mathcal{F}_1, B \in \mathcal{F}_2\}$, then $\sigma\langle\mathcal{C}\rangle = \mathcal{F}_1 \times \mathcal{F}_2$. Clearly, $\mathcal{F}_1 \times \mathcal{F}_2 \subset \mathcal{P}(\mathbb{N} \times \mathbb{N})$ since the power set of $\mathbb{N} \times \mathbb{N}$ is the largest possible sigma algebra on $\mathbb{N}^2$. Secondly, for any $D \in \mathcal{P}(\mathbb{N} \times \mathbb{N})$, as $\mathbb{N} \times \mathbb{N}$ is countable, $D = \cup_{i \geq 1}\{(a_i, b_i)\}, a_i, b_i \in \mathbb{N}$. Then $D \in \sigma\langle\mathcal{C}\rangle$ follows immediately from the property, closed under countable union, of a sigma algebra. Therefore, $\mathcal{F}_1 \times \mathcal{F}_2 = \mathcal{P}(\mathbb{N} \times \mathbb{N})$.

(ii) Since both $\{(n, n) : n \in \mathbb{N}\}$ and $\{(n, n+1) : n \in \mathbb{N}\}$ are in $\mathcal{F}_1 \times \mathcal{F}_2$, f is a $\mathcal{F}_1 \times \mathcal{F}_2$ measurable function.

(iii) Clearly, $\int_{\Omega_1 \times \Omega_2} |f(\omega_1, \omega_2)| d\mu_1 \times \mu_2 = \sum_{n=1}^{\infty} \mu_1 \times \mu_2(\{(n, n)\}) + \sum_{n=1}^{\infty} \mu_1 \times \mu_2(\{(n, n+1)\}) = \infty$, f is not integrable w.r.t. $\mu_1 \times \mu_2$.

(iv) For each $\omega_1 \in \mathbb{N}$, $\int_{\Omega_2} f(\omega_1, \omega_2) d\mu_2(\omega_2) = [I_{\{(n,n):n\in\mathbb{N}\}}(\omega_1, \omega_1) - I_{\{(n,n+1):n\in\mathbb{N}\}}(\omega_1, \omega_1 + 1)] = 0$, so

$$\int_{\Omega_1} \left[\int_{\Omega_2} f(\omega_1, \omega_2) d\mu_2(\omega_2)\right] d\mu_1(\omega_1) = 0.$$

Similarly $\int_{\Omega_1} f(\omega_1, 1) d\mu_1(\omega_1) = 1$ and $\int_{\Omega_1} f(\omega_1, \omega_2) d\mu_1(\omega_1) = 0$ for all $\omega_2 \geq 2$, so

$$\int_{\Omega_2} \left[\int_{\Omega_1} f(\omega_1, \omega_2) d\mu_1(\omega_1)\right] d\mu_2(\omega_2) = \mu_2(\{1\}) = 1.$$